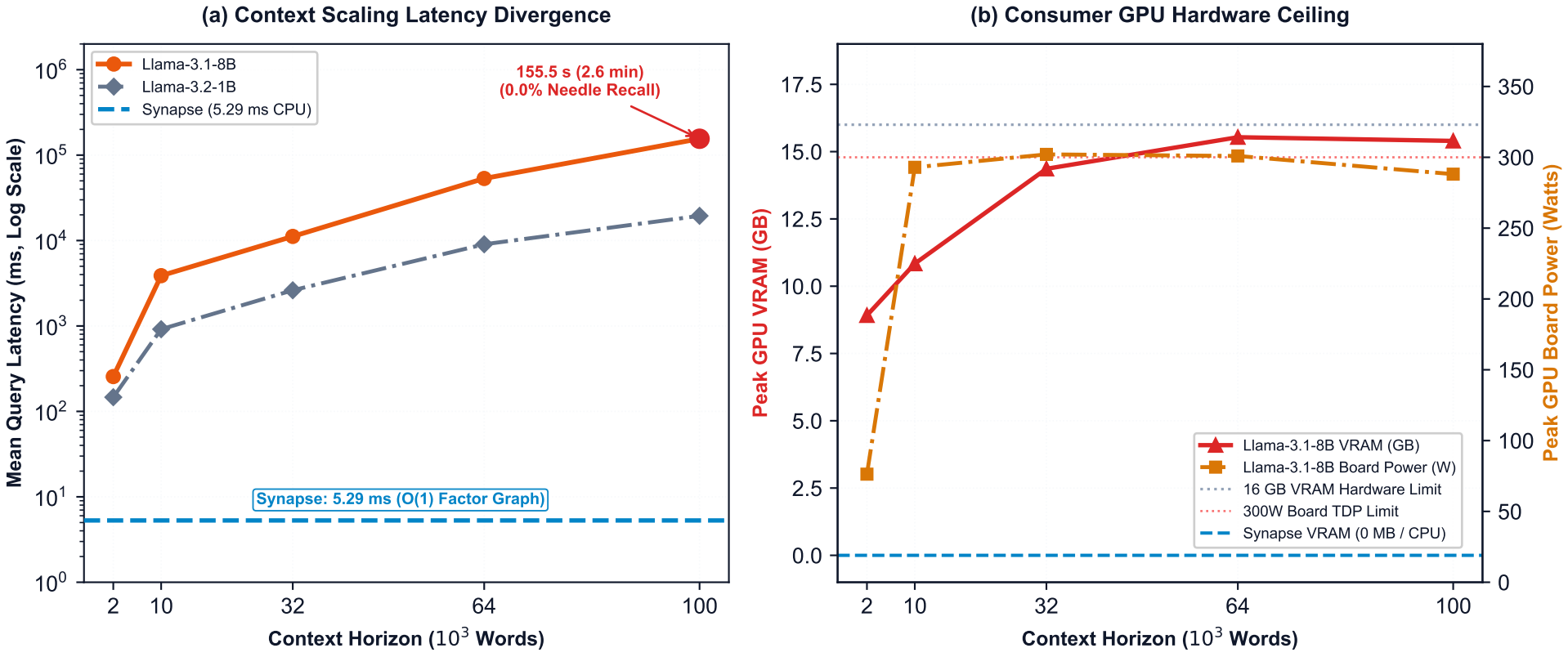
(a) Context Scaling Latency Divergence
Llama-3.1-8B
Llama-3.2-1B
Synapse (5.29 ms CPU)
155.5 s (2.6 min)
(0.0% Needle Recall)
Synapse: 5.29 ms (O(1) Factor Graph)
Mean Query Latency (ms, Log Scale)
Context Horizon ($10^3$ Words)
(b) Consumer GPU Hardware Ceiling
Peak GPU VRAM (GB)
Peak GPU Board Power (Watts)
Llama-3.1-8B VRAM (GB)
Llama-3.1-8B Board Power (W)
16 GB VRAM Hardware Limit
300W Board TDP Limit
Synapse VRAM (0 MB / CPU)
Context Horizon ($10^3$ Words)